CIE AS PHYSICS (9702) • QUESTION PAPER

8.3 Interference

Syllabus Section: 8. Superposition **Paper Type:** Multiple Choice (Paper 1) **Total:** 16 Questions (16 Marks)

Question 1 Easy [1 mark]

Two fringes were created on a screen in a double-slit experiment; the distances were too small to measure.

Which of the following would increase the distance between the fringes?

A increasing the frequency of the light source

B increasing the distance between the light source and the slits

C increasing the distance between the slits and the screen

D increasing the distance between the slits

Question 2 Easy [1 mark]

A student investigates the interference of light using two identical green LEDs.

Which statement would explain why the investigation will not work?

A the light waves from the sources have a range of wavelengths

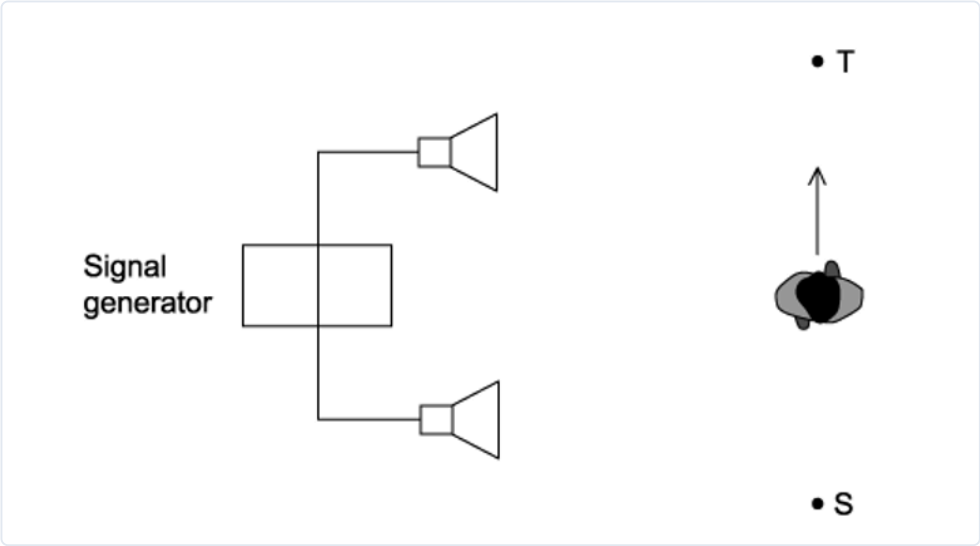
B the light waves from the sources are not monochromatic

C the light waves from the sources are not coherent

D the light waves from the sources do not have the same amplitude

Question 3 Easy [1 mark]

Two loudspeakers are connected to a signal generator, as shown in the diagram.



As the student walks between S to T she notices that the loudness of the sound increases then decreases repeatedly.

Why does the loudness change?

A polarisation of the sound waves

B reflection of the sound waves

C diffraction of the sound waves

D interference of the sound waves

Question 4 Easy [1 mark]

Fig 1.1 shows two waves

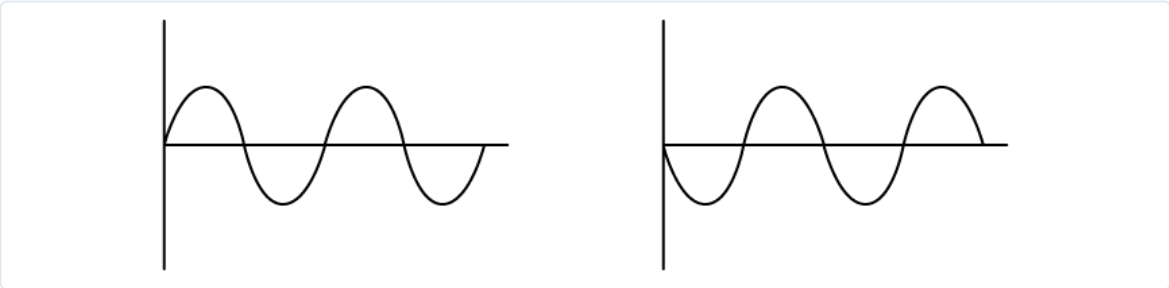


Fig 1.1

Which of the following statements is true?

A The waves are coherent

B The waves are incoherent

C The waves are in phase

D When added together, the waves would cause constructive interference

Question 5 Easy [1 mark]

Monochromatic light of wavelength λ is incident upon a double slit and forms an interference pattern on a screen.

What is a possible path difference for a minima that appears on the screen?

A 0

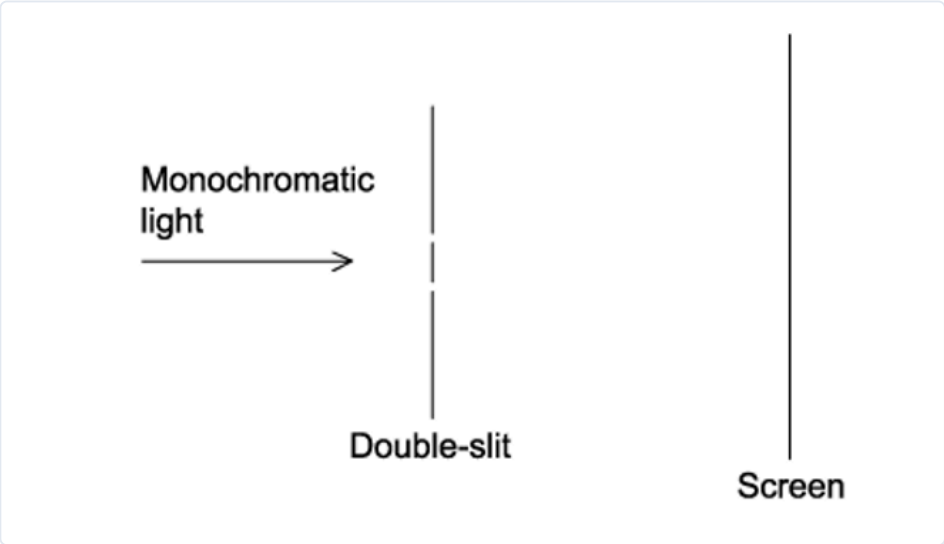
B $\frac{3}{2}\lambda$

C $3 \quad \lambda$

D $5 \quad \lambda$

Question 6 Medium [1 mark]

The effect of a double slit on monochromatic light is observed by set up in the diagram



Interference fringes are seen on the screen.

Which of the following changes would increase the distance between the adjacent fringes?

A decrease the width of each slit

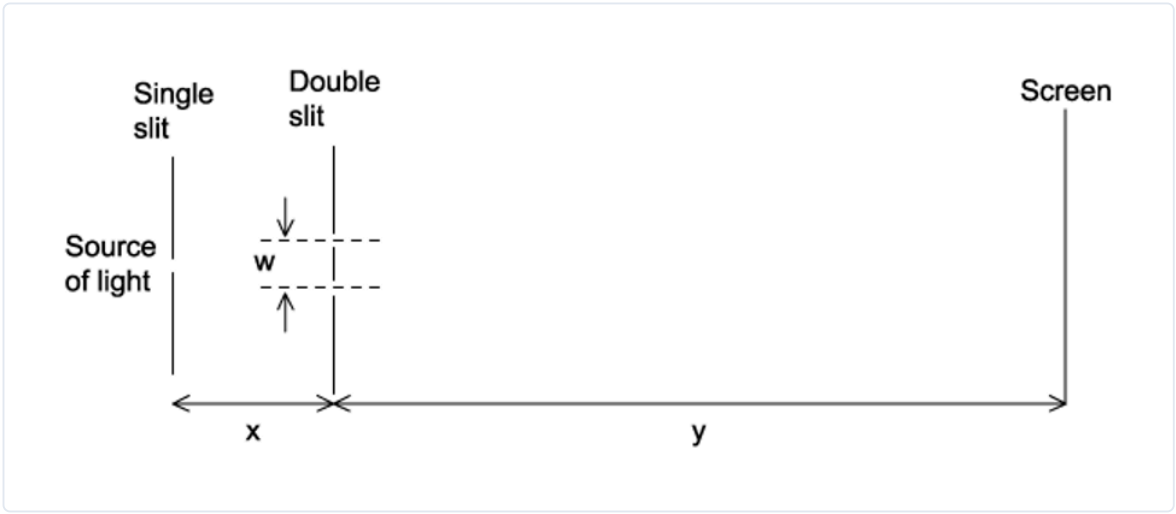
B decrease the distance between the two slits

C use light of a higher frequency

D move the screen closer to the double-slit

Question 7 Medium [1 mark]

The equipment was set up as in the diagram to investigate the interference of a double slit.



Which of the following changes would increase the fringe spacing?

A decreasing the wavelength of the light

B decreasing distance *y*

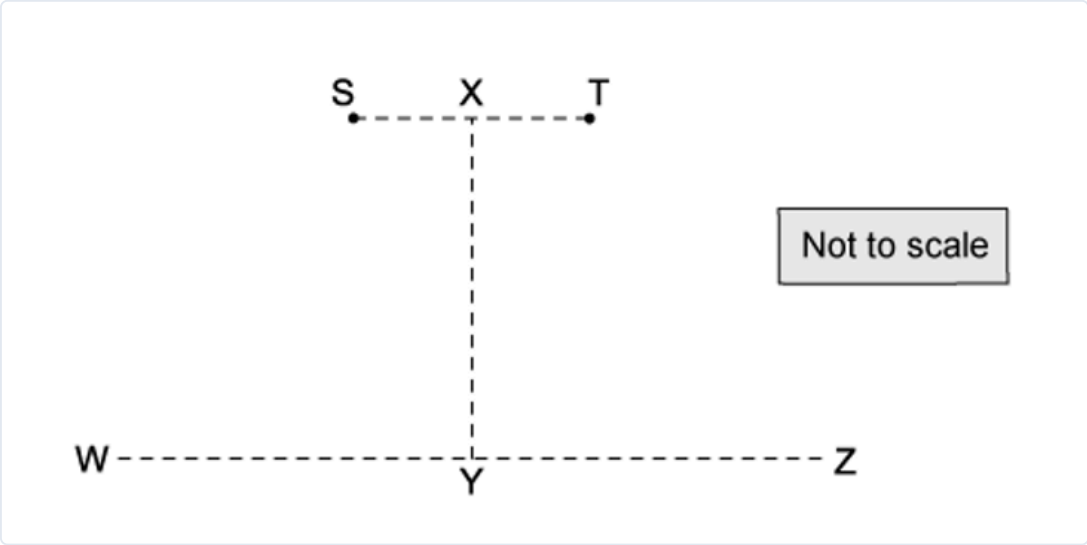
C decreasing distance *x*

D decreasing the distance *w*

Question 8 Medium [1 mark]

A student is investigating the interference of waves. They set up coherent waves produced at points S and T that travel outwards in all directions.

The line XY is halfway between S and T and perpendicular to the joining line S and T. The distance XY is much greater than the distance ST.



Which line or lines would an interference pattern be seen on?

A WZ only

B XY only

C Both XY and WZ

D Neither XY or WZ

Question 9 Medium [1 mark]

Certain circumstances will produce visible interference fringes from two light sources.

Which of the following conditions will enable visible interference fringes to be seen?

A using one light which is polarised at right angles to light from the other source

B using incoherent sources

C using a blue light source

D using sources from which the light does not overlap

Question 10 Medium [1 mark]

Monochromatic light produces interference fringes with a separation of x on a screen a distance D from a pair of slits with separation a.

The student doubles a and D.

What would be the new fringe separation?

A $2x$

B $4x$

C x

D $\frac{x}{2}$

Question 11 Medium [1 mark]

Red light with a wavelength of 650 nm is used in a Young's double-slit experiment. The distance between the double slit and the screen is 2.5 m and the separation of the slits is 0.80 mm.

What would be the fringe separation on the screen?

A 0.13 mm

B 0.21 mm

C 1.3 mm

D 2.0 mm

Question 12 Hard [1 mark]

Yellow light with a wavelength of 600 nm is used in an experiment to show interference fringes produced by double-slit interference. The fringe separation is 4.0 mm and the separation of the slits is 0.60 mm.

What would be the distance between the double slit and the screen?

A 4.0 m

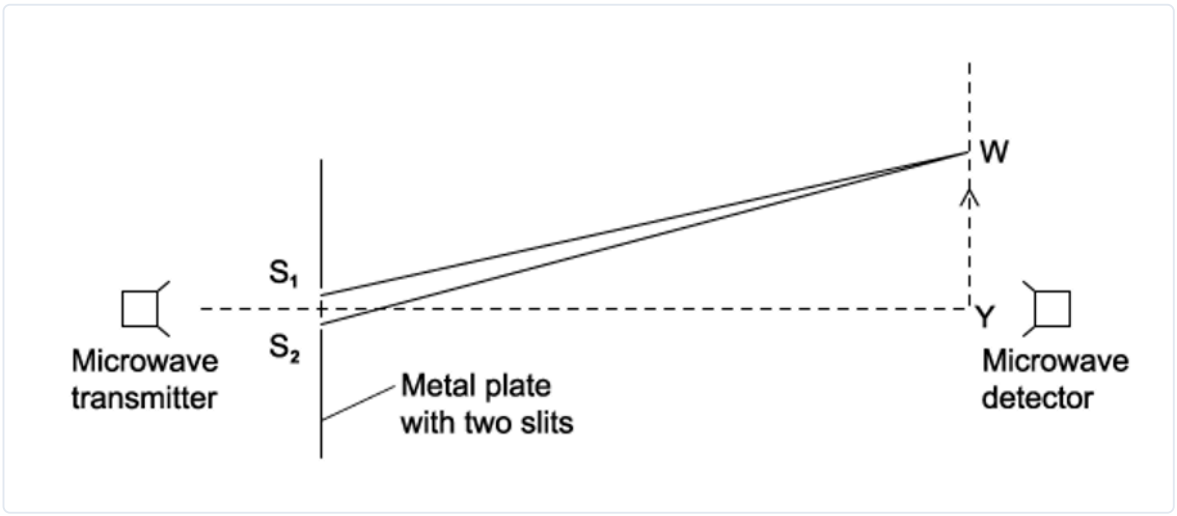
B 2.5 m

C 0.4 m

D 0.25 m

Question 13 Hard [1 mark]

An experiment was set up to investigate two-source interference. Microwaves of wavelength λ are passed through S_1 and S_2.



The detector is moved in the direction of the arrow from Y to W. The signal detected is decreased to zero at point W and then starts to increase again as it continues past W.

Which of the following would be the equation to determine the position of point W?

A $S_2W - S_1W = \frac{\lambda}{2}$

B $S_2W - S_1W = \lambda$

C $YW = \frac{\lambda}{2}$

D $YW = \lambda$

Question 14 Hard [1 mark]

Yellow light with a wavelength of 600 nm is used in an experiment to show interference fringes produced by double-slit interference. The fringe separation is 4.0 mm on the screen.

What would be the fringe spacing if the wavelength is changed to 400 nm and the separation of the slits is doubled?

A 1.3 mm

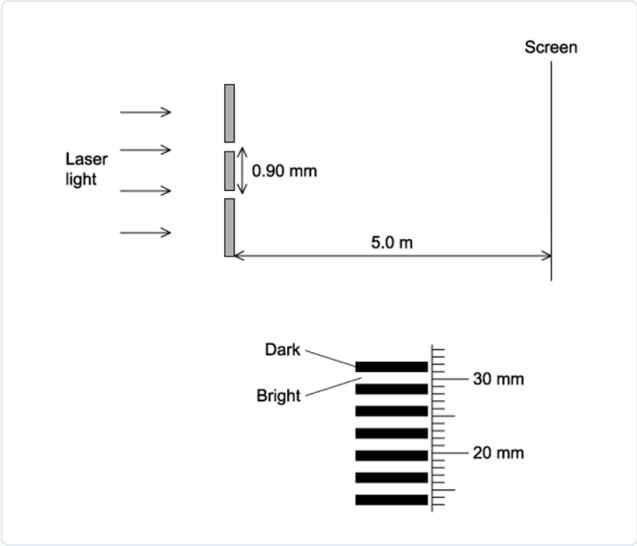
B 3.0 mm

C 5.3 mm

D 12.0 mm

Question 15 Hard [1 mark]

A student carried out an experiment on Young's slits using a laser light. There is also a section of the pattern produced with a ruler placed next to it.



What is the wavelength of the light?

A 540 nm

B 3.4 μm

C 480 nm

D 3.2 μm

Question 16 Hard [1 mark]

A beam of monochromatic light with wavelength λ is shone through a double slit at distance D away from a screen. An interference pattern is produced where the distance between the fringes is x.

What would the distance between the double slit and the screen need to be for the fringe distance to remain x when a light beam of wavelength $\frac{3}{4}\lambda$ is used instead?

A $\frac{1}{4}D$

B $\frac{1}{3}D$

C $\frac{3}{4}D$

D $\frac{4}{3}D$